

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PCE 204 — Past Questions & Materials
by Renance

Course: PCE 204
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — PCE 204 - 2024/2025 - Second Semester - PETROCHEMISTRY

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
EFFURUN

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COLLEGE OF ENGINEERIN AND TECHNOLOGY

DEPARTMENT OF PETROCHEMICAL ENGINEERING

SECOND SEMESTER EXAMINATION; 2024/2025 ACADEMIC SESSION

COURSE TITTLE:] PETROCHEMISTRY Credit Units:1

COURSE CODE: PCE 204 Time Allowed:1; hrs

INSTRUCTIONS: Answer ANY THREE (3) Questions. All questions carry equal marks

QUESTION ONE

aid (a) Briefly explain the following terms as applied to Thermodynamics

i, A System ii State iii. Closed System iv. Adiabatic system

(b) From the First Law of Thermodynamics, write the expression for the thermodynamic variable, the

enthalpy, H and for the enthalpy change in a constant pressure process.

(c) A rigid tank of 0.1m^3 contains a mixture of saturated steam and saturated water at a pressure of 2bar. By volume, the mixture is 10% liquid. How much heat must be added so that the tank contain only saturated steam? What will be the pressure in the tank?

5 . (QUESTION TWO

)ts ad { (a) Briefly highlight the differences between the First and Second laws of Thermodynamics.

(b) Explain the terms, i. molar conductivity ii. ionic conductivity.

(c) Explain the gaps in the first and second laws of thermodynamics and how they are filled by the third law of thermodynamics.

be QUESTION THREE

(a) Briefly explain the mode of action of a catalyst.

(b) Distinguish between homogenous and heterogeneous catalysts. -

(c) What is the salient difference between a base catalyst and acid catalyst? x

; "U 4 ey (d) Explain the features of a colloid. : ve

QUESTION FOUR

(a) Explain the basic differences between primary raw materials and secondary raw materials with Us Qa Ww specific examples.

(b) Differentiate between associated and non-associated natural gas

(c) State five non hydrocarbon constituent of natural gas.

x

(d) What is produce water? Yl ee We |

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(a) What are the properties of natural gas?

(b) Explain the physical adsorption process of natural gas treatment

Ny What is the relevance of chromatographic analysis in improved natural gas liquid recovery?

d) Describe the process of formation of petroleum IP

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